

14-3b The Long Run: Market Supply with Entry and Exit

Now consider what happens when firms are able to enter and exit the market. Let's suppose that everyone has access to the same technology for producing the good and access to the same markets to buy the inputs for production. Therefore, all current and potential firms have the same cost curves.

Decisions about entry and exit in a market of this type depend on the incentives facing the owners of existing firms and the entrepreneurs who could start new firms. If firms already in the market are profitable, then new firms will have an incentive to enter the market. This entry will expand the number of firms, increase the quantity of the good supplied, and drive down prices and profits. Conversely, if firms in the market are making losses, then some existing firms will exit the market. Their exit will reduce the number of firms, decrease the quantity of the good supplied, and drive up prices and profits. *At the end of this process of entry and exit, firms that remain in the market must be making zero economic profit.*

Recall that we can write a firm's profit as

$$\text{Profit} = (P - ATC) \times Q.$$

This equation shows that an operating firm has zero profit if and only if the price of the good equals the average total cost of producing that good. If price is above average total cost, profit is positive, which encourages new firms to enter. If price is less than average total cost, profit is negative, which encourages some firms to exit. *The process of entry and exit ends only when price and average total cost are driven to equality.*

This analysis has a surprising implication. We noted earlier in the chapter that competitive firms maximize profits by choosing a quantity at which price equals marginal cost. We just noted that free entry and exit force price to equal average total cost. But if price is to equal both marginal cost and average total cost, these two measures of cost must equal each other. Marginal cost and average total cost are equal, however, only when the firm is operating at the minimum of average total cost. Recall from the preceding chapter that the level of production with lowest average total cost is called the firm's *efficient scale*. Therefore, *in the long-run equilibrium of a competitive market with free entry and exit, firms must be operating at their efficient scale.*

Panel (a) of [Figure 7](#) shows a firm in such a long-run equilibrium. In this figure, price P equals marginal cost MC , so the firm is maximizing profit. Price also equals average total cost ATC , so profit is zero. New firms have no incentive to enter the market, and existing firms have no incentive to leave the market.

Figure 7

Long-Run Market Supply

In the long run, firms will enter or exit the market until profit is driven to zero. As a result, price equals the minimum of average total cost, as shown in panel (a). The number of firms adjusts to ensure that all demand is satisfied at this price. The long-run market supply curve is horizontal at this price, as shown in panel (b).

From this analysis of firm behavior, we can determine the long-run supply curve for the market. In a market with free entry and exit, there is only one price consistent with zero profit—the minimum of average total cost. As a result, the long-run market supply curve must be horizontal at this price, as illustrated by the perfectly elastic supply curve in panel (b) of [Figure 7](#). Any price above this level would generate profits, leading to entry and an increase in the total quantity supplied. Any price below this level would generate losses, leading to exit and a decrease in the total quantity supplied. Eventually, the number of firms in the market adjusts so that price equals the minimum of average total cost, and there are enough firms to satisfy all the demand at this price.